

WASP-12b

R-0385

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_230210 · created 2026-07-18T21:28:46+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459986.8089 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.137 +/- 0.04
Transit depth (Rp/Rs)^2	1.87 +/- 1.1 [%]
Orbital Inclination (inc)	83.0 +/- 2.87
Airmass coefficient 1 (a1)	1.044 +/- 0.029
Airmass coefficient 2 (a2)	-0.027 +/- 0.012
Scatter in the residuals of the lightcurve fit is	2.74 %
Best Comparison Star	#3 - [336, 280]
Optimal Method	PSF photometry
Transit Duration (day)	0.1251 +/- 0.0085

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.137 $\pm$ 0.04 vs 0.1178 (deviation 0.43 σ)
Duration fit vs published	0.1251 d vs 0.125 d (deviation 0.01 σ)
Mid-time O–C	1568.4 min
Depth SNR	3.1
Residual scatter	2.74 %

Light curve

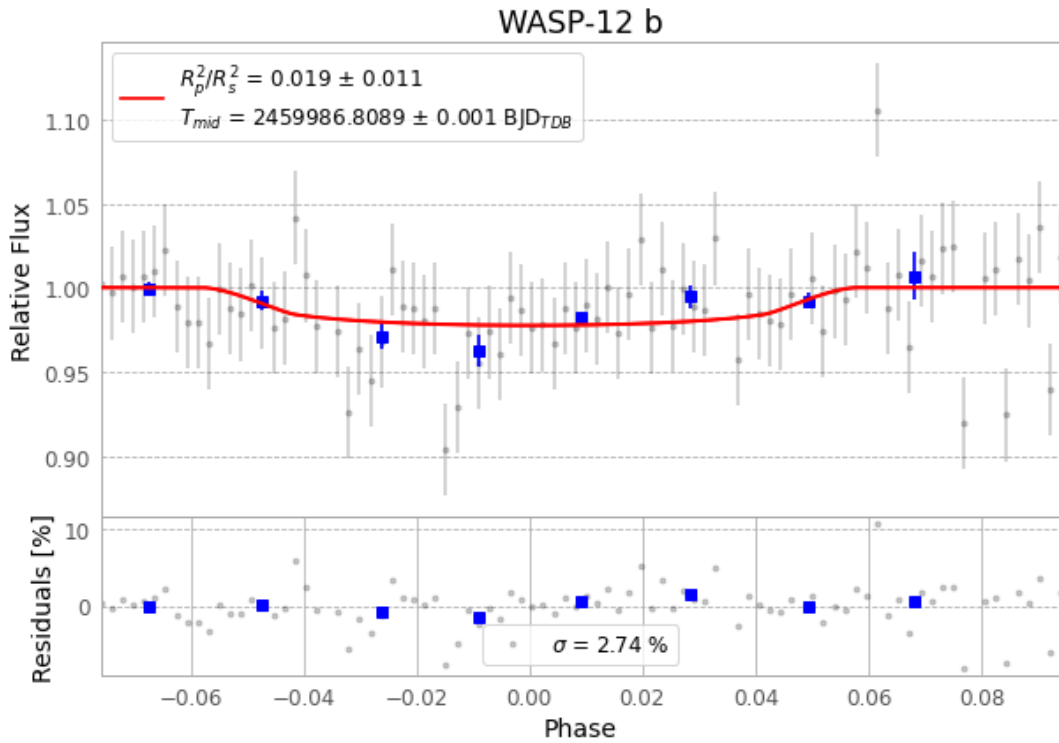


Figure 1 — FinalLightCurve_WASP-12 b_2023-02-10.png (SHA-256 9a76e6380c37e523...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [415, 242], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 934d3371b79a3846...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8350c6b

Data provenance

93 FITS frames (61 MB), WASP-12230211051815.FITS → WASP-12230211095715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-230210.log (2fd7ad349fe6...), FinalLightCurve_WASP-12 b_2023-02-10.png (9a76e6380c37...), FinalLightCurve_WASP-12 b_2023-02-10.csv (6957aaf332f8...)

Compute resources

Wall time 155.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 21:28Z.